Performance Testing

Date	28 September 2026
Team ID	SWTID-2026-3161
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing and Spike Testing
Target Module / API	ServiceNow Table API (GET) on the table protected by the script-controlled ACL
Test Environment	Cloud – ServiceNow Personal Developer Instance (PDI)
Test Date	28 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal load: authorized users read records that satisfy the field-value condition	10	60	Records returned; avg response time under 2 sec
2	Unauthorized users request restricted records (field value not allowed)	10	60	Restricted records not returned; no server errors
3	Peak load: mixed authorized and unauthorized users	50	120	Correct access decisions; error rate under 1%
4	Spike test: sudden jump in users	100	60	System stays stable; max response time under 5 sec

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	ACL script adds very little delay.
2	Response Time (Max)	< 5 seconds	3.4 seconds	Pass	Highest value seen during the spike test.
3	Throughput (Req/sec)	> 10 req/sec	14.6 req/sec	Pass	Stable across scenarios 1 to 3.

4	Error Rate	< 1%	0.4%	Pass	Only a few timeouts during the spike test.
5	CPU Utilization	< 80%	46%	Pass	Checked on the ServiceNow instance stats page.
6	Memory Utilization	< 80%	52%	Pass	No memory growth during the test.

Step 4: Observations & Analysis

Key Findings:

The script-controlled ACL correctly allowed and denied access in all scenarios. Average and maximum response times stayed within the targets under normal, peak and spike load.

Bottlenecks Identified:

Response time increased slightly during the spike test because the ACL script ran a role check and a field lookup for every record.

Optimization Steps Taken:

Access-check logic was moved to a reusable Script Include, duplicate role checks were removed, and the ACL condition was simplified so the script runs only when needed.

Step 5: Screenshots / Evidence

Evidence of the performance testing tool results (JMeter test configuration and summary report) is given below:

Test Configuration:

Item	Details
Test Plan	ACL_Performance_Test.jmx (JMeter Thread Group per scenario)
Sampler	HTTP Request – GET /api/now/table/<target_table>
Authentication	Basic Auth with test users (authorized and unauthorized roles)
Ramp-up Time	10 seconds for all scenarios
Listeners Used	Summary Report, Aggregate Report, Response Time Graph
Assertions	Response code 200; restricted records must not appear for unauthorized users

JMeter Summary Report (Aggregate):

Scenario	Samples	Average (ms)	Min (ms)	Max (ms)	Error %	Throughput (req/sec)
1. Normal load (10 users)	880	1050	420	2100	0.0%	14.7
2. Unauthorized access (10 users)	870	980	390	1900	0.0%	14.5
3. Peak load (50 users)	1750	1220	450	2900	0.3%	14.6
4. Spike test (100 users)	870	1650	480	3400	1.4%	14.6
TOTAL	4370	1224	390	3400	0.4%	14.6